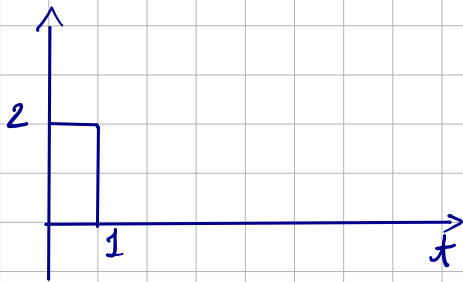


WEEK-4 GRADED ASSIGNMENT

Q.1]



$$f(t) = 2 \quad 0 < t \leq 1$$

$$f(t) = 0 \quad t > 1$$

$$\therefore \mathcal{L} f(t) = \int_{0^-}^{\infty} f(t) \cdot e^{-st} dt$$

$$= \int_{0^-}^1 f(t) \cdot e^{-st} dt + \int_1^{\infty} f(t) \cdot e^{-st} dt$$

$$= -2 \cdot \left[\frac{e^{-st}}{s} \right]_{0^-}^1 + 0$$

$$= \frac{2}{s} [1 - e^{-s}]$$

$$= \frac{2 - 2e^{-s}}{s}$$

Q.2] $\frac{dy}{dt} + ay = e^{-bt}$ $y(0) = 0$

Soln:- If $I_f = e^{\int a dt} = e^{at}$

$$\therefore y e^{at} = \int e^{-bt} \cdot e^{at} dt$$

$$y \cdot e^{at} = \int e^{(a-b)t} dt$$

$$y \cdot e^{at} = \frac{e^{(a-b)t}}{a-b} + c$$

$$y = \frac{e^{at} \cdot e^{-bt}}{(a-b) e^{at}} + c \cdot e^{-at}$$

$$y = \frac{e^{-bt}}{(a-b)} + c e^{-at}$$

we know $y(0) = 0$

$$0 = \frac{1}{(a-b)} + c$$

$$c = \frac{1}{b-a}$$

$$\therefore y = \frac{e^{-bt}}{(a-b)} - \frac{e^{-at}}{(a-b)}$$

$$y = \frac{e^{-(b+a)t}}{(a-b)}$$

Method-2

$$s \cdot Y(s) + a Y(s) = \frac{e^{-(b+s)t}}{-(b+s)} \int_0^\infty$$

$$y(s) [s+a] = \frac{1}{(b+s)}$$

$$y(s) = \frac{1}{(s+a)(s+b)}$$

Answer

$$\mathcal{L} y(t) = \int_0^\infty \frac{e^{-(b+a)t}}{(a-b)} \cdot e^{-st} dt$$

$$= \int_0^\infty \frac{e^{-(b+a+s)t}}{(a-b)} dt$$

$$= \frac{1}{(a-b)} \cdot \frac{e^{-(b+a+s)t}}{(b+a+s)} \Big|_0^\infty$$

$$= \frac{1}{(a-b)} \left[\frac{1}{(b+a+s)} - 0 \right]$$

$$= \frac{1}{(a-b)(a+b+s)}$$

Q3]

$$\text{Soln:- } \mathcal{L} f(t) = \frac{(s+3)}{(s+1)(s+2)}$$

$$\frac{(s+3)}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$\frac{(s+3)}{(s+2)} = A + \frac{B(s+1)}{(s+2)}$$

$$0 = A + \frac{B(-2)}{-1}$$

$$A = -2B$$

$$\boxed{A = 2}$$

$$\frac{(s+3)}{(s+1)} = \frac{(s+2)A}{(s+1)} + B$$

$$\frac{1}{-1} = B$$

$$\boxed{B = -1}$$

$$\frac{(s+3)}{(s+1)(s+2)} = \frac{2}{(s+1)} - \frac{1}{(s+2)}$$

$$\therefore \mathcal{L}^{-1} \frac{(s+3)}{(s+1)(s+2)} = \mathcal{L}^{-1} \frac{2}{(s+1)} - \mathcal{L}^{-1} \frac{1}{(s+2)}$$

$$= 2e^{-t} - e^{-2t}$$

$$f(t) = 2e^{-t} - e^{-2t}$$

$$f(0) = 2 - 1$$

$$f(0) = 1$$

Q.4]

$$\mathcal{L} \frac{1}{(s+1)} = e^{-1t}$$

$$= e^{-t}$$

Proof: By verification.

$$f(t) = e^{-t}$$

$$\mathcal{L} f(t) = \int_0^{\infty} e^{-t} \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{-(1+s)t} dt$$

$$= - \frac{e^{-(1+s)t}}{(1+s)} \Big|_0^{\infty}$$

$$= \frac{1}{1+s}$$

Q.5]

Soln:-

$$\mathcal{L} 6u(t-1) + 5u(t-5)$$

$$= 6 \int_1^{\infty} u(t-1) e^{-st} dt + 5 \int_5^{\infty} u(t-5) \cdot e^{-st} dt$$

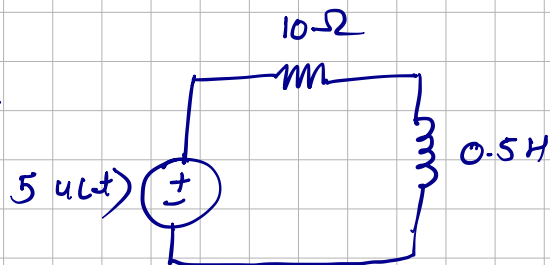
$$= 6 \cdot \frac{e^{-st}}{s} \Big|_1^{\infty} + 5 \cdot \frac{e^{-st}}{s} \Big|_5^{\infty}$$

$$= \frac{6}{s} \cdot [e^{-s}] + \frac{5}{s} e^{-5s}$$

$$= \frac{6e^{-s} + 5e^{-5s}}{s}$$

Q.6]

Soln:-



$$5 u(t) = 10 + L \frac{di}{dt}$$

$$\mathcal{L} 5 u(t) = \mathcal{L} 10i + \mathcal{L} L \cdot \frac{di}{dt}$$

$$\begin{aligned} \frac{5}{s} &= 10 F(i) + L \cdot \int_0^{\infty} \frac{di}{dt} \cdot e^{-st} dt \\ &\quad + L \left(e^{-st} \cdot i \int_0^{\infty} + \int_0^{\infty} s \cdot e^{-st} \cdot i dt \right) \\ &\quad + L \cdot (-i(0^-)) + Ls \cdot F(i) \end{aligned}$$

$$\frac{5}{s} = 10 F(i) + L (s \cdot F(i) - i(0^-))$$

$$i(0^-) = 0 \quad L = 0.5$$

$$\frac{5}{s} = 10 F(i) + 0.5 s F(i)$$

$$\frac{5}{s} = F(i) [10 + 0.5s]$$

$$F(i) = \frac{5}{s(10 + 0.5s)}$$

Q.7) $f(t) = \begin{cases} t-1 & 1 < t < 2 \\ 3-t & 2 < t < 3 \\ 0 & \text{otherwise} \end{cases}$

Soln:-

$$\mathcal{L} f(t) = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$= \int_0^1 0 \cdot e^{-st} dt + \int_1^2 (t-1) \cdot e^{-st} dt + \int_2^3 (3-t) \cdot e^{-st} dt$$

$$+ \int_3^{\infty} 0 \cdot e^{-st} dt$$

$$= \int_1^2 (t-1) \cdot e^{-st} dt + \int_2^3 (3-t) \cdot e^{-st} dt$$

$$= \int_1^2 t \cdot e^{-st} dt - \int_1^2 e^{-st} dt + \int_2^3 3 \cdot e^{-st} dt - \int_2^3 t \cdot e^{-st} dt$$

$$= \int_1^2 \frac{d}{ds} e^{-st} dt + (e^{-2s} - e^{-s}) - 3 \cdot (e^{-3s} - e^{-2s}) - \int_2^3 \frac{d}{ds} e^{-st} dt$$

$$= \frac{d}{ds} \int_1^2 e^{-st} dt + \frac{d}{ds} \int_2^3 e^{-st} dt + (e^{-2s} - e^{-s}) - 3(e^{-3s} - e^{-2s})$$

$$= \frac{d}{ds} \left[\frac{1}{s} (e^{-s} - e^{-2s}) \right] + \frac{d}{ds} \left[\frac{1}{s} (e^{-3s} - e^{-2s}) \right]$$

$$= \frac{s \cdot (-e^{-s} + 2e^{-2s}) - (e^{-s} - e^{-2s})}{s^2} + \frac{s \cdot (-3e^{-3s} + 2e^{-2s}) - (e^{-3s} - e^{-2s})}{s^2}$$

$$\therefore \frac{2e^{-2s} - e^{-s}}{s} - \frac{(e^{-s} - e^{-2s})}{s^2} + \frac{2e^{-2s} - 3e^{-3s}}{s} - \frac{(e^{-3s} - e^{-2s})}{s^2}$$

$$\begin{aligned}
 & -\frac{e^{-s}}{s} - \frac{3e^{-3s}}{s} - \frac{e^{-s}}{s^2} + \frac{e^{-2s}}{s^2} - \frac{e^{-3s}}{s^2} + \frac{e^{-2s}}{s^2} \\
 & = \frac{e^{-s}}{s^2} - \frac{2e^{-2s}}{s^2} + \frac{e^{-3s}}{s^2} \\
 & = \frac{1}{s^2} (e^{-s} - 2e^{-2s} + e^{-3s})
 \end{aligned}$$

Q.8) $f(t) = \cos^3(t)$

we know $\cos(t) = \frac{e^{jt} + e^{-jt}}{2}$

$$\begin{aligned}
 \cos^3(t) &= \left(\frac{e^{jt} + e^{-jt}}{2} \right)^3 \\
 &= \frac{1}{8} (e^{3jt} + e^{-3jt} + 3e^{2jt} \cdot e^{-jt} + 3e^{-2jt} \cdot e^{jt})
 \end{aligned}$$

$$\mathcal{L} \cos^3 t = \mathcal{L} \frac{1}{8} (e^{3jt} + e^{-3jt} + 3e^{jt} + 3e^{-jt})$$

$$= \frac{1}{8} \int_0^{\infty} (e^{3jt} + e^{-3jt} + 3e^{jt} + 3e^{-jt}) \cdot e^{-st} dt$$

$$= \frac{1}{8} \int_0^{\infty} e^{(3j-s)t} dt + \int_0^{\infty} e^{-(3j+s)t} dt + \int_0^{\infty} 3 \cdot e^{(j-s)t} dt + \int_0^{\infty} 3 \cdot e^{-(j+s)t} dt$$

$$= \frac{1}{8} \left[\left. \frac{e^{(3j-s)t}}{(3j-s)} \right|_0^{\infty} - \left. \frac{e^{-(3j+s)t}}{(3j+s)} \right|_0^{\infty} + \frac{3 \cdot e^{(j-s)t}}{(j-s)} \right]_0^{\infty} - \frac{3 \cdot e^{-(j+s)t}}{(j+s)} \right]_0^{\infty}$$

$$= \frac{1}{8} \left[\frac{1}{s-3j} + \frac{1}{(3j+s)} - \frac{3}{(j-s)} + \frac{3}{(j+s)} \right]$$

$$= \frac{1}{8} \left[\frac{s+3j + s-3j}{s^2+9} + \frac{3(s+j) + 3(s-j)}{s^2+1} \right]$$

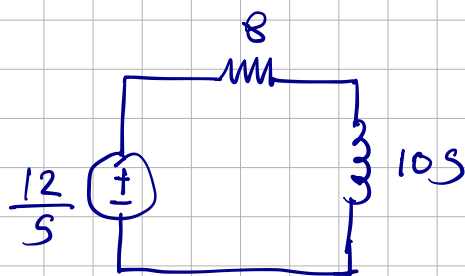
$$= \frac{1}{8} \left[\left(\frac{2s}{s^2+9} \right) + \left(\frac{6s}{s^2+1} \right) \right]$$

$$= \frac{0.25s}{s^2+9} + \frac{0.75s}{s^2+1}$$

Q.9)

Soln:-

$$I(s) = \frac{12}{s(8+10s)}$$



$$\therefore R = 8$$

$$L = 10$$

Q.10)

Soln:-

$$l = 10$$

∞ out of (finite.)

Q.11)

Soln:-

$$\frac{1}{s(s^2+9)} = \frac{1}{s(s+j3)(s-j3)} = \frac{A}{s} + \frac{B}{(s+j3)} + \frac{C}{(s-j3)}$$

$$\frac{1}{(s+j3)(s-j3)} = A + \frac{B(s)}{(s+j3)} + \frac{C(s)}{(s-j3)}$$

$$\frac{1}{9} = A$$

$$\frac{1}{s(s-j3)} = \frac{1}{9}(s+j3) + B + \frac{C(s+j3)}{(s-j3)}$$

$$\frac{1}{-3j(-6j)} = B$$

$$\frac{1}{-18} = B$$

$$\frac{1}{s(s+j3)} = \frac{A(s-j3)}{s} + \frac{B(s-j3)}{(s+j3)} + \frac{C}{s}$$

$$\frac{1}{(j3)(6j)} = C$$

$$\frac{1}{-18} = C$$

$$\therefore F(s) = \frac{1}{9s} - \frac{1}{18(s+j3)} - \frac{1}{18(s-j3)}$$

$$\mathcal{L}^{-1} F(s) = \mathcal{L}^{-1} \frac{1}{9s} - \mathcal{L}^{-1} \frac{1}{18(s+j3)} - \mathcal{L}^{-1} \frac{1}{18(s-j3)}$$

$$= \frac{1}{9} u(t) - \frac{1}{18} e^{-3jt} - \frac{1}{18} e^{j3t}$$

$$= \frac{1}{9} u(t) - \frac{1}{18} [\cos 3t - j \sin(3t)] - \frac{1}{18} [\cos 3t + j \sin 3t]$$

$$= \frac{1}{9} u(t) - \frac{1}{18} \cos(3t) + \frac{j \sin(3t)}{18} - \frac{\cos 3t}{18} - \frac{j \sin 3t}{18}$$

$$= \frac{1}{9} u(t) - \frac{2}{18} \cos(3t)$$

$$= \frac{1}{9} - \frac{\cos(3t)}{9}$$

$$= \frac{1}{9} (1 - \cos(3t))$$